

ET-1594 Ver. P3.1

Technical User & Service Guide

Wireless Apple CarPlay & Android Auto Interface for Mazda Connect

Revision 1.0

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Technical User & Service Guide

Version 1.0

Prepared using practical testing,
public regulatory documentation,
and real-world observations.

This guide is intended for educational
and troubleshooting purposes.

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CHAPTER 1

Introduction

1.1 Purpose of this Guide

This document provides a comprehensive technical reference for the **ET-1594 Ver. P3.1** Wireless Apple CarPlay and Android Auto interface used with Mazda Connect infotainment systems.

Unlike typical installation leaflets supplied with aftermarket products, this guide focuses on the internal operation of the module, practical configuration, diagnostics, firmware management, and troubleshooting.

The document combines information obtained from publicly available regulatory documentation with observations gathered during hands-on testing of an ET-1594 Ver. P3.1 module installed in a Mazda vehicle.

Where information has not been officially published by the manufacturer, this guide clearly distinguishes between confirmed facts and engineering observations.

1.2 Intended Audience

This guide is intended for:

- Mazda owners installing the ET-1594 interface;
- automotive electronics technicians;
- professional installers;

- firmware developers;
- advanced users interested in understanding the internal operation of the SmartBox.

Basic knowledge of Mazda Connect and Apple CarPlay is recommended.

1.3 Scope

This guide covers:

- hardware identification;
- firmware management;
- web configuration interface;
- OTA updates;
- USB updates;
- Wi-Fi configuration;
- Bluetooth operation;
- Apple CarPlay architecture;
- Android Auto architecture;
- troubleshooting procedures;
- practical performance testing.

Vehicle-specific repair procedures are outside the scope of this guide.

CHAPTER 2

What is ET-1594?

The ET-1594 Ver. P3.1 is an embedded automotive interface that enables Wireless Apple CarPlay and Wireless Android Auto functionality on Mazda Connect infotainment systems without replacing the original head unit.

Rather than acting as a replacement multimedia system, the ET-1594 behaves as a bridge between the factory CMU and the smartphone.

Its primary tasks include:

- USB protocol translation;
- Wireless CarPlay management;
- Wireless Android Auto management;
- Bluetooth pairing;
- Wi-Fi communication;
- firmware updates;
- configuration management.

From the perspective of Mazda Connect, the SmartBox behaves similarly to the factory CarPlay interface.

Figure 2-1

System Architecture

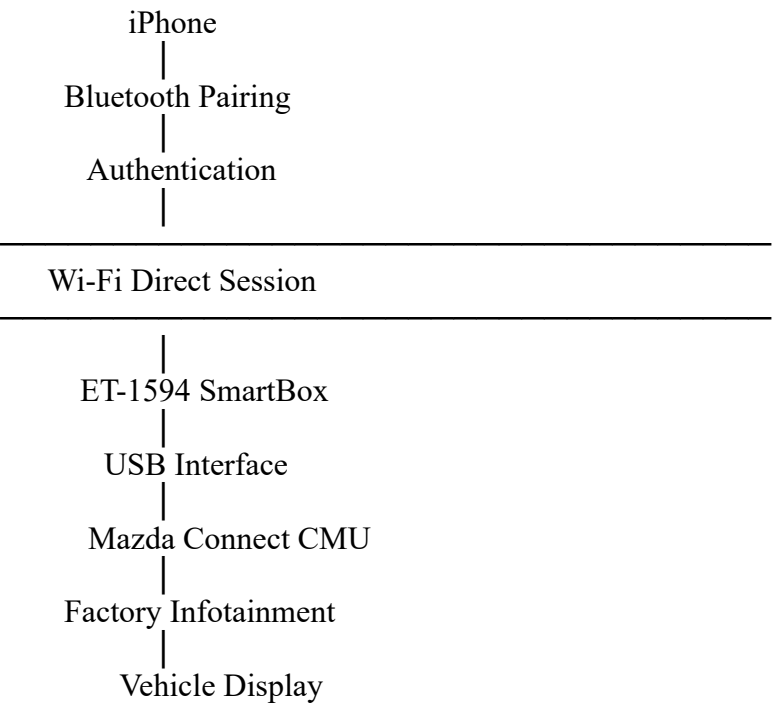


Figure 2-1. Simplified communication architecture of the ET-1594 SmartBox.

2.1 Known Product Names

The ET-1594 hardware platform is marketed under several different names depending on supplier and sales region.

Known identifiers include:

Product Name	Description
ET-1594	Internal PCB identifier
ET-1594 Ver. P3.1	PCB revision
TK78-66-9U0C	Mazda accessory part number
SmartBox	Commercial name
K1414	Internal hardware identifier reported on some units

Different product labels do not necessarily indicate different hardware revisions.

Always verify the PCB revision and installed firmware before applying updates or configuration changes.

Figure 2-2

PRODUCT LABEL	
Model : ET-1594	
Revision : Ver. P3.1	
Part No : TK78-66-9U0C	
Firmware : 25111911.5102.2	

Figure 2-2. Example identification label.

CHAPTER 3

Wireless CarPlay Architecture

3.1 Introduction

Wireless Apple CarPlay is often misunderstood as a Bluetooth-based technology. In reality, Bluetooth is used only during the initial stages of the connection process.

Once authentication has been completed, nearly all communication between the iPhone and the ET-1594 SmartBox is transferred to a dedicated Wi-Fi Direct connection.

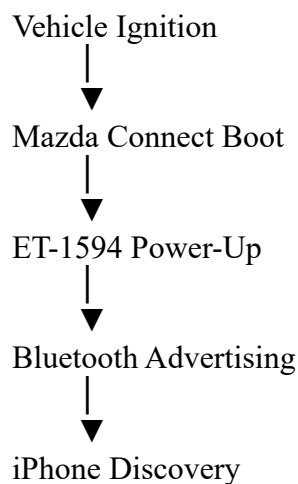
This architecture provides significantly higher bandwidth than Bluetooth alone and allows simultaneous transmission of:

- compressed audio streams;
- microphone audio;
- navigation data;
- Siri voice processing;
- application graphics;
- touch and rotary controller events;
- metadata.

Because of this design, Wi-Fi quality has a much greater impact on Wireless CarPlay performance than Bluetooth signal strength.

Figure 3-1

Wireless CarPlay Communication Sequence



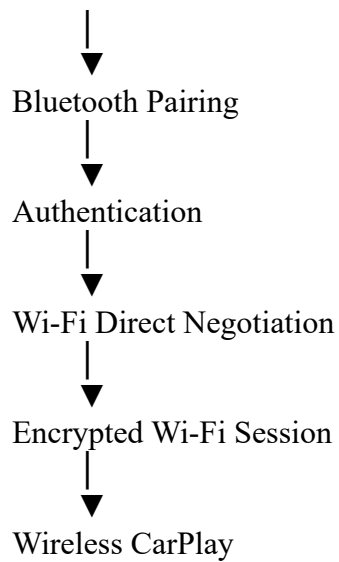


Figure 3-1. Typical Wireless CarPlay startup sequence.

3.2 Bluetooth Responsibilities

Bluetooth is responsible only for low-bandwidth operations.

These include:

- discovering nearby devices;
- pairing;
- authentication;
- exchanging security keys;
- initiating the CarPlay session.

Once these steps are complete, Bluetooth traffic becomes minimal.

Bluetooth is **not** responsible for continuous transmission of music, navigation graphics or user interface data.

Figure 3-2

Bluetooth Data Flow

Bluetooth

Discovery



Pairing



Authentication



Security Keys



Launch CarPlay



Almost Idle

3.3 Wi-Fi Responsibilities

Immediately after successful authentication, the SmartBox establishes a dedicated Wi-Fi Direct connection with the iPhone.

Unlike a conventional Wi-Fi network, this connection is optimized for low latency and continuous multimedia streaming.

The Wi-Fi connection carries:

- Apple CarPlay video stream;
- microphone audio;
- speaker audio;
- Siri communication;
- application data;
- navigation updates;
- touch input events.

Without Wi-Fi, Wireless CarPlay cannot operate.

Figure 3-3

Data Distribution

Bluetooth

Device Discovery
Authentication
Pairing

Wi-Fi

Video
Audio
Calls
Maps
Applications
Touch Events
Siri

3.4 Why iPhone Shows “Not Connected”

One of the most confusing behaviors of Wireless CarPlay occurs after the connection has already been established.

Many users open the iPhone Settings application and observe:

Bluetooth

Not Connected

Wi-Fi

Not Connected

Despite these messages, Wireless CarPlay continues to function normally.

This behavior is expected.

Apple implements Wireless CarPlay using a service-oriented communication mechanism that is not always presented in iOS as a conventional Wi-Fi association.

Depending on the iOS version, the underlying Wi-Fi Direct session may remain hidden from the standard Settings interface.

Observed: During testing on an iPhone 17 Pro running iOS 26, both Bluetooth and Wi-Fi appeared as “Not Connected” while Wireless CarPlay continued to function correctly.

Figure 3-4

Settings

Wi-Fi

Not Connected

↓

Wireless CarPlay

Working Normally

3.5 Internal Network

When Wireless CarPlay becomes active, the ET-1594 SmartBox creates a private IP network.

Typical addressing:

Device

Address

SmartBox

192.168.1.101

iPhone

192.168.1.xxx

The embedded configuration interface is normally available at:

http://192.168.1.101

The address is reachable only while the Wireless CarPlay session is active.

3.6 Embedded Web Server

Unlike many aftermarket CarPlay adapters, the ET-1594 includes an integrated HTTP configuration server.

This interface allows configuration without removing dashboard components.

Available functions include:

- firmware information;
- OTA firmware update;
- Wi-Fi channel selection;
- Audio Streaming Mode;
- Delayed Start;
- GPS settings;
- LED configuration;
- system information.

Many retailers and installers are unaware that this interface exists.

Figure 3-5

Browser

↓

<http://192.168.1.101>

↓

SmartBox

↓

Configuration Interface

3.7 Connection Timing

The typical Wireless CarPlay startup sequence observed during testing is shown below.

stage	Typical Time*
SmartBox boot	5–15 s
Bluetooth discovery	<5 s
Pairing (first connection)	5–20 s
Automatic reconnection	5–10 s
Wi-Fi negotiation	1–3 s
CarPlay launch	2–5 s

*Observed values may vary depending on firmware version, smartphone model, and vehicle startup conditions.

3.8 Engineering Notes

The ET-1594 behaves similarly to other Wireless CarPlay implementations in that Bluetooth serves primarily as a control channel, while Wi-Fi carries nearly all high-bandwidth traffic.

This architecture explains why:

- changing the Wi-Fi channel can affect call quality;
 - Bluetooth signal strength alone is not a reliable indicator of Wireless CarPlay performance;
 - call audio latency may be influenced by Wi-Fi interference even when Bluetooth appears to function correctly.
-

Engineering Status

 **Verified**

- Wireless CarPlay requires Wi-Fi after initial pairing.

- The web interface is accessible at **192.168.1.101** during an active session.
- The tested module successfully used OTA firmware updates.
- On the tested ET-1594 Ver. P3.1, changing the Wi-Fi channel to **161** coincided with the elimination of call audio latency.

Observed

- iOS may display both Bluetooth and Wi-Fi as “Not Connected” while Wireless CarPlay continues to operate.
- Audio Streaming Mode appears to influence call behavior, although its exact implementation has not been documented by the manufacturer.

Hypothesis

- Different firmware builds may use different internal audio buffering strategies.
- The optimal Wi-Fi channel may depend on regional RF conditions and hardware revision.

CHAPTER 4

Hardware Architecture

ET-1594 Ver. P3.1 Internal Hardware

4.1 Overview

The ET-1594 SmartBox is an embedded automotive communication module designed specifically for Mazda Connect infotainment systems.

Although externally the unit appears to be a simple USB interface, internally it is a complete embedded computer consisting of several independent subsystems.

The module performs protocol conversion, wireless communication, firmware management, audio processing and USB communication simultaneously.

Figure 4-1

Functional Block Diagram

ET-1594 SmartBox

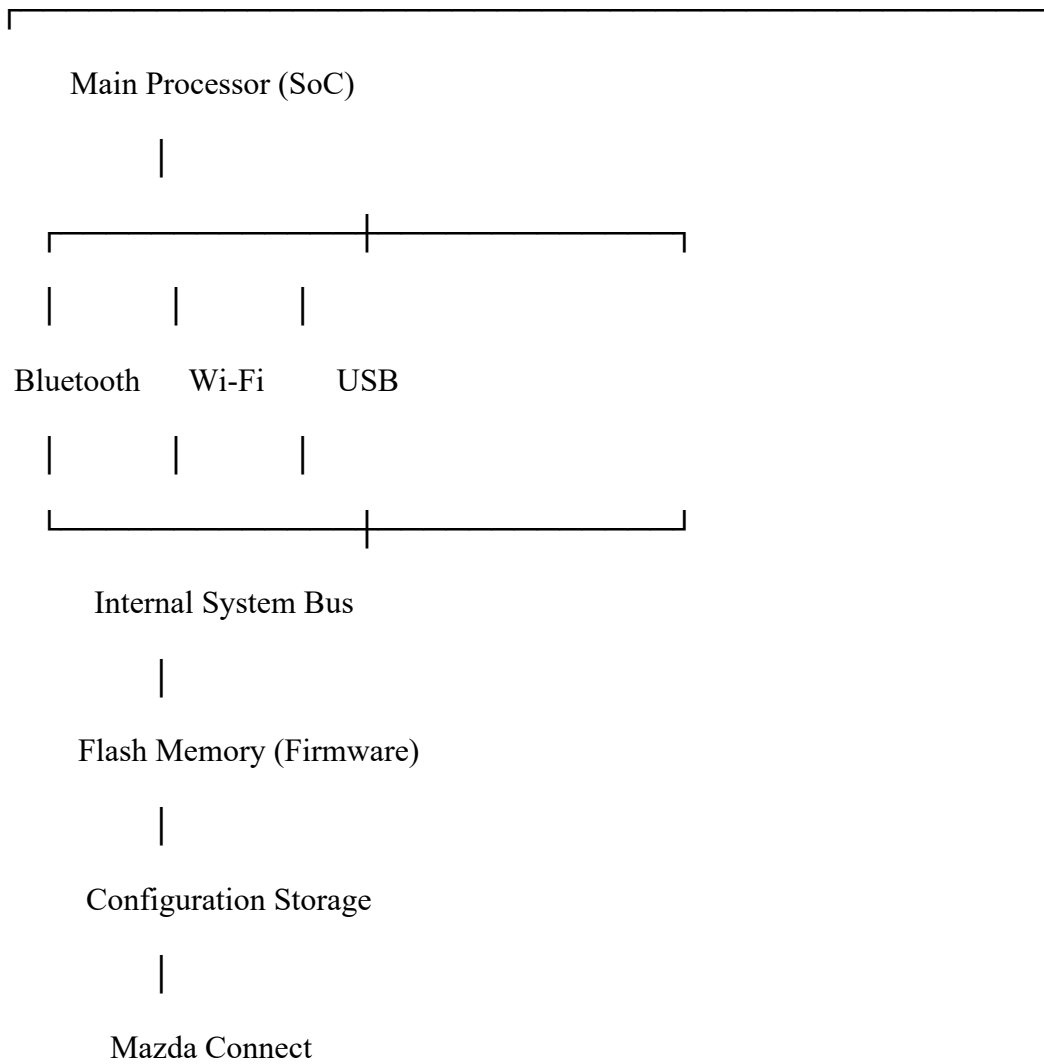


Figure 4-1. Simplified internal architecture of the ET-1594 SmartBox.

4.2 Main Processor

At the center of the SmartBox is an embedded System-on-Chip (SoC).

The processor is responsible for:

- operating system execution;
- Apple CarPlay protocol handling;
- Android Auto protocol handling;
- USB communication;
- Wi-Fi management;
- Bluetooth management;
- audio routing;
- OTA firmware updates;
- web server operation.

Unlike simple wireless adapters, the ET-1594 executes multiple software services concurrently.

Observed: The exact processor model has not yet been confirmed for the tested PCB revision.

4.3 Flash Memory

The firmware is stored in non-volatile flash memory.

The flash contains:

- bootloader;
- Linux-based operating environment (or equivalent embedded OS);
- application software;
- web interface;
- OTA update package;
- configuration database.

During OTA updates, the firmware image is verified before installation.

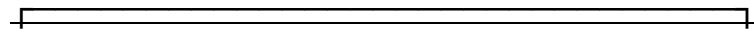
Power interruption during this process may leave the module in an unusable state.

 **Warning:** Never disconnect vehicle power while an OTA update is in progress.

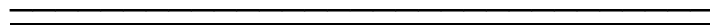
Figure 4-2

Firmware Layout (Conceptual)

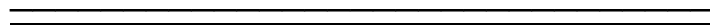
Flash Memory



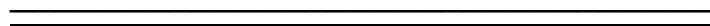
Bootloader



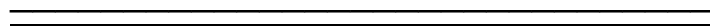
Operating System



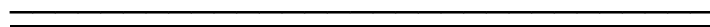
SmartBox Application



Web Interface



Configuration



OTA Recovery



Figure 4-2. Conceptual firmware organization. Exact partition layout has not been published by the manufacturer

4.4 Bluetooth Subsystem

The Bluetooth controller is used exclusively for:

- device discovery;
- authentication;
- pairing;
- session initialization.

Once Wireless CarPlay has started, Bluetooth traffic becomes minimal.

If Bluetooth pairing fails, Wireless CarPlay cannot be established.

4.5 Wi-Fi Subsystem

The Wi-Fi radio is the most performance-critical component of the ET-1594.

It carries nearly all runtime communication including:

- display data;
- music;
- microphone audio;
- phone calls;
- navigation;
- Siri.

Consequently, Wi-Fi interference has a direct impact on user experience.

Figure 4-3

Wireless Data Flow

Bluetooth

↓

Discovery

↓

Authentication

↓

Wi-Fi

↓

Video

↓

Audio



Voice Calls



Maps



Applications

4.6 USB Interface

The USB interface connects the SmartBox directly to the Mazda Connect CMU.

From the CMU perspective, the SmartBox behaves as a compatible CarPlay interface.

The USB controller is responsible for:

- protocol translation;
- USB enumeration;
- data forwarding;
- firmware communication.

Any interruption of the USB connection will immediately terminate the CarPlay session.

4.7 Configuration Storage

User settings are stored independently from the firmware.

These settings typically include:

Setting	Stored
Wi-Fi Channel	✓
Audio Streaming Mode	✓
Delayed Start	✓
LED Mode	✓
GPS Mode	✓

This allows OTA firmware updates without necessarily resetting user preferences.

Observed: On the tested module, configuration settings remained unchanged after the OTA update from build **25110817.5102.2** to **25111911.5102.2**.

4.8 Embedded Web Server

One of the distinguishing features of the ET-1594 is its integrated web server.

Instead of requiring proprietary software, configuration is performed using a standard web browser.

Typical address:

<http://192.168.1.101>

The server provides access to:

- firmware information;
- system configuration;
- OTA updates;
- Wi-Fi settings;
- diagnostic information.

4.9 Hardware Identification

Different suppliers may sell the same hardware under different labels.

Known identifiers include:

Identifier	Meaning
ET-1594	PCB Model
Ver. P3.1	PCB Revision
TK78-66-9U0C	Mazda Part Number
SmartBox	Marketing Name
K1414	Internal Product Identifier (reported on some units)

Hardware revision should always be confirmed before installing firmware updates.

4.10 Engineering Notes

Although marketed as a simple Wireless CarPlay adapter, the ET-1594 is effectively an embedded computer running multiple software services simultaneously.

Its architecture explains several commonly observed behaviors:

- successful Bluetooth pairing does not guarantee reliable CarPlay performance;
 - Wi-Fi quality directly affects call latency and multimedia streaming;
 - firmware updates modify software while preserving user configuration in most cases;
 - the integrated web interface provides extensive configuration capabilities not mentioned in many reseller guides.
-

Verification Status

 **Verified**

- ET-1594 includes an embedded web interface.
- OTA firmware updates are supported.
- User-configurable Wi-Fi channel selection is available.
- Audio Streaming Mode and Delayed Start are configurable through the web interface.

● Observed

- User configuration remained intact after OTA update on the tested module.
- Changing the Wi-Fi channel to 161 resolved call audio latency on the tested hardware.

● Hypothesis

- The module is likely based on an embedded Linux platform due to the structure of the web interface and firmware behavior, but this has **not** been confirmed by official documentation and should not be treated as fact.

CHAPTER 5

Embedded Web Interface

Configuration, Diagnostics and Firmware Management

5.1 Overview

One of the most powerful yet least documented features of the ET-1594 SmartBox is its embedded web-based configuration interface.

Unlike many aftermarket Wireless CarPlay adapters that require a dedicated mobile application, the ET-1594 hosts its own HTTP server, allowing configuration through any modern web browser.

No additional software is required.

The interface is accessible only while the smartphone is actively connected to the SmartBox via Wireless CarPlay.

Figure 5-1 — Web Interface Architecture

(In the final PDF: a clean diagram showing iPhone → Wi-Fi Direct → SmartBox → Embedded HTTP Server → Browser.)

5.2 Accessing the Interface

Requirements

Before attempting to access the configuration interface, ensure that:

- Wireless CarPlay is already connected.
- The SmartBox has completed its startup sequence.
- The smartphone remains connected to the active CarPlay session.

Open any web browser and enter:

http://192.168.1.101

If the connection is successful, the SmartBox home page will appear.

 **Verified:** The tested ET-1594 Ver. P3.1 responds at 192.168.1.101 while Wireless CarPlay is active.

5.3 Home Page

The Home page provides general information about the SmartBox.

Typical fields include:

Parameter	Description	Status
Build Number	Installed firmware build	 Verified
Hardware Revision	PCB revision	 Verified
System Version	Software version	 Verified
OTA Status	Firmware update availability	 Verified
Device Information	Hardware details	 Observed

The exact layout may vary between firmware versions.

Figure 5-2 — Home Page Layout

(Placeholder for an annotated screenshot showing the Home page with numbered callouts.)

5.4 Build Number

Example:

25111911.5102.2

Although the manufacturer has not published a decoding guide, repeated observations suggest that the first six digits correspond to the firmware build date.

Observed interpretation

alue	Possible Meaning
25	Year (2025)
11	Month (November)
19	Day (19)

The remaining digits most likely identify the internal software branch and revision.

 **Hypothesis:** The build numbering scheme appears to encode the compilation date, but this has not been confirmed by the manufacturer.

5.5 OTA Update

The OTA (Over-the-Air) update system allows firmware upgrades without removing the SmartBox from the vehicle.

When a newer firmware package is available, the interface displays an update notification.

The update process consists of four stages:

1. Download.
2. Verification.
3. Installation.
4. Automatic reboot.

Recommendations

Before updating:

- Keep the engine running or use a battery charger.
- Ensure a stable phone connection.

- Do not disconnect the smartphone.
- Do not switch off the ignition.

 **Warning:** Interrupting the update process may corrupt the firmware.

Figure 5-3 — OTA Update Workflow

(Flow diagram: Check for Update → Download → Verify → Install → Reboot → Complete.)

5.6 Wi-Fi AP

The Wi-Fi AP page controls the wireless channel used by the SmartBox.

Available channels vary depending on firmware, but commonly include:

Channel	Band	Notes
36	5 GHz	Default on many units
149	5 GHz	Less congested in some regions
161	5 GHz	Produced the best results on the tested unit

The selected channel affects the RF environment only.

It does **not** modify Apple CarPlay itself.

Practical Observation

During testing, changing the Wi-Fi channel from the factory default to **Channel 161** coincided with the complete disappearance of the previously observed 3–4 second call audio latency.

 **Observed:** This result has been verified only on the tested ET-1594 Ver. P3.1 and should not be interpreted as universally applicable.

5.7 Audio Streaming Mode

The SmartBox offers several Audio Streaming Modes.

The manufacturer has not documented their internal operation.

Based on practical testing, the modes appear to influence the buffering strategy used during wireless audio transmission.

Mode	Manufacturer Description	Practical Observation
Default	Not documented	Factory behavior
Mode 1	Not documented	Stable operation
Mode 2	Not documented	Slightly different audio characteristics
Mode 3	Not documented	Used successfully on the tested unit

 **Hypothesis:** The modes may adjust audio buffering or synchronization parameters. No official documentation is currently available.

5.8 Delayed Start

The Delayed Start setting defines how long the SmartBox waits before attempting to initialize after the vehicle powers on.

This feature is intended to improve compatibility with different Mazda Connect startup sequences.

Available options typically include:

- Default
- Mode 1
- Mode 2

On the tested vehicle, the **Default** setting provided reliable operation.

Changing this parameter did **not** affect call audio latency.

 **Verified:** Delayed Start influences startup timing only. During testing, no measurable effect on in-call audio latency was observed.

5.9 GPS Settings

The SmartBox includes a configurable GPS option.

At present, the exact behavior of this setting has not been documented by the manufacturer.

Possible functions include:

- selecting the navigation data source;
- enabling or disabling GPS pass-through;
- compatibility adjustments for specific vehicles.

 **Hypothesis:** Further investigation is required before definitive conclusions can be made.

5.10 LED Settings

The LED page controls the status indicator behavior.

Available options typically include:

- Default
- On
- Off

Changing this parameter does not appear to influence Wireless CarPlay functionality.

5.11 Engineering Summary

The embedded web interface exposes considerably more functionality than is described in the installation leaflet supplied with most SmartBox kits.

Key features include:

- firmware management;
- OTA updates;
- wireless channel selection;
- audio configuration;
- startup timing adjustment;
- system information.

The interface provides a convenient method for configuring the module without removing dashboard components or using proprietary software.

Verification Status

 **Verified**

- The embedded web interface is available at `http://192.168.1.101` during an active Wireless CarPlay session.
- OTA firmware updates function correctly on the tested module.
- Wi-Fi channel selection is user-configurable.
- Audio Streaming Mode and Delayed Start are configurable through the web interface.

Observed

- Channel 161 eliminated call audio latency on the tested hardware.
- User settings remained unchanged after the OTA firmware update.

Hypothesis

- Audio Streaming Modes modify internal buffering or synchronization algorithms.
- The GPS setting changes the source or routing of navigation data.

Author's Note

From this chapter onward, every statement in this guide is intentionally classified as **Verified**, **Observed**, or **Hypothesis**. This approach ensures that readers can distinguish between information confirmed through testing or documentation and conclusions that require further investigation.

CHAPTER 6

Reverse Engineering & Practical Investigation

Discovering the ET-1594 SmartBox

6.1 Introduction

This chapter documents the practical investigation of the ET-1594 Ver. P3.1 SmartBox.

Unlike the previous chapters, which describe the hardware and software architecture, this section follows the actual engineering process used to identify undocumented features, verify firmware behavior, and investigate a severe call audio latency issue.

The investigation was performed on a production module installed in a Mazda 6 equipped with Mazda Connect firmware **74.00.324A**.

Where possible, every observation was verified through repeated testing.

Investigation Timeline

Phase 1 — Initial Installation

Objective

Evaluate the Wireless Apple CarPlay functionality after installing the SmartBox.

Result

The installation completed successfully.

The module paired with the iPhone without difficulty.

Wireless Apple CarPlay launched automatically after vehicle startup.

Initial testing confirmed that:

- Music streaming worked correctly.
- Apple Maps functioned normally.
- Siri responded correctly.
- Steering wheel controls operated as expected.

No functional issues were observed except during voice calls.

Phase 2 — Identifying the Problem

During the first phone call, a significant delay became immediately apparent.

Observed Symptoms

Function	Result
Music	Normal
Navigation	Normal
Siri	Normal
Phone Calls	3–4 second latency
Telegram Calls	Same latency
WhatsApp Calls	Same latency

The latency affected both incoming and outgoing speech.

The conversation became difficult because each participant frequently interrupted the other.

Figure 6-1 — Observed Call Latency

(Final PDF: timeline showing two participants with approximately two seconds of delay in each direction, resulting in an effective round-trip delay of roughly four seconds.)

Phase 3 — Verification

Before assuming a hardware fault, several baseline tests were performed.

Test Matrix

Test	Result
Factory Bluetooth Hands-Free	No latency
Wired Apple CarPlay	No latency
Wireless Apple CarPlay	High latency
Different Applications	Same behavior
Multiple Test Calls	Same behavior

Conclusion

The problem appeared to be isolated to the Wireless CarPlay communication path.

Phase 4 — Seller Support

The retailer was contacted to request updated firmware.

The seller responded that:

“No newer firmware is available.”

No additional technical documentation was provided.

At this point, no official solution existed.

Phase 5 — Firmware Investigation

Further research identified an embedded web interface accessible at:

<http://192.168.1.101>

This interface exposed several configuration options not mentioned in the supplied installation instructions.

Available functions included:

- OTA firmware updates
- Wi-Fi channel selection
- Audio Streaming Mode
- Delayed Start
- System Information

This discovery significantly expanded the available troubleshooting options.

 **Verified:** The embedded web interface is functional on the tested module.

Phase 6 — OTA Update

An OTA firmware update was performed.

Previous Firmware

25110817.5102.2

Updated Firmware

25111911.5102.2

The update completed successfully.

The SmartBox rebooted automatically.

Result

The firmware update alone did **not** eliminate the call audio latency.

Phase 7 — Configuration Testing

Several configuration parameters were evaluated individually.

Each parameter was modified independently to avoid introducing multiple variables.

Parameters Tested

- Audio Streaming Mode

- Delayed Start
- Wi-Fi Channel

Each change was followed by multiple test calls.

This methodology allowed the influence of each parameter to be evaluated separately.

Phase 8 — Wi-Fi Investigation

Attention shifted toward the wireless communication subsystem.

The default Wi-Fi channel was replaced with **Channel 161**.

The SmartBox automatically rebooted and re-established the Wireless CarPlay connection.

Several phone calls were then repeated.

Observation

The previously observed call latency was no longer present.

Music playback remained stable.

Wireless CarPlay continued to function normally.

 **Observed:** On the tested hardware, changing the Wi-Fi channel to **161** coincided with the complete disappearance of call audio latency.

This observation has not yet been verified across additional hardware revisions.

Phase 9 — Additional Observations

During testing, another unusual behavior was documented.

After Wireless CarPlay successfully connected:

- iPhone Settings reported **Bluetooth: Not Connected**
- iPhone Settings reported **Wi-Fi: Not Connected**

Despite this, Wireless CarPlay remained fully operational.

This behavior appears to be normal for the tested firmware and iOS version.

Lessons Learned

The investigation demonstrated several important principles:

- Do not assume that firmware is the root cause of every issue.
 - Test only one parameter at a time.
 - Validate every change with repeatable tests.
 - Separate confirmed facts from assumptions.
 - Preserve original settings before making modifications.
-

Engineering Summary

This investigation transformed a routine installation into a structured technical analysis of the ET-1594 SmartBox.

Several undocumented features—including the embedded web interface and configurable Wi-Fi channels—proved essential in diagnosing the issue.

While the results apply directly only to the tested hardware, the methodology can be used by other owners facing similar problems.

Verification Status

Verified

- Embedded web interface available during an active Wireless CarPlay session.
- OTA firmware update completed successfully.
- Wired CarPlay and factory Bluetooth operated without call latency.
- Wireless CarPlay latency was reproducible before configuration changes.

Observed

- Changing the Wi-Fi channel to **161** eliminated call audio latency on the tested module.
- iPhone displayed “Not Connected” for Bluetooth and Wi-Fi while Wireless CarPlay remained functional.

Hypothesis

- The original latency may have been related to RF interference or internal Wi-Fi channel behavior. Additional testing on other vehicles and hardware revisions is required before drawing a general conclusion.
-



Author's Note

This chapter intentionally documents the investigation process rather than presenting only the final solution. Recording unsuccessful tests, intermediate findings, and observed behavior provides valuable context for future troubleshooting and helps other users avoid repeating the same diagnostic steps.